

Python for Data Science

Complete Roadmap: Beginner to Expert

Industry-Relevant Edition 2026

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1. Prerequisites & Setup

No prior programming experience is required to begin this roadmap. However, the following foundational skills will help you move faster and get more out of each phase.

1.1 Recommended Background Knowledge

- Basic math: arithmetic, fractions, percentages, and simple algebra
- Familiarity with spreadsheets (Excel or Google Sheets is helpful but not mandatory)
- A curious mindset and willingness to experiment and debug

1.2 Technical Setup

- Install Python 3.11+ from python.org or use Anaconda Distribution (recommended for data science)
- Install VS Code or JupyterLab as your development environment
- Create a free GitHub account to track and share your projects
- Install Git for version control from git-scm.com
- Create accounts on Kaggle and Google Colab for datasets and free cloud compute

1.3 Essential Tools Overview

Tool	Purpose	Difficulty
Anaconda	Python distribution with pre-installed data science libraries	Easy
JupyterLab	Interactive notebooks for exploration and visualization	Easy
VS Code	Code editor for scripts and larger projects	Easy
Git & GitHub	Version control and portfolio hosting	Moderate
Google Colab	Free cloud-based Jupyter environment with GPU access	Easy
Kaggle	Datasets, competitions, and learning notebooks	Easy

PHASE 1 — BEGINNER

This phase builds your programming foundation. The goal is to write clean Python code confidently, understand data structures, and take your first steps with data manipulation. Do not rush — every concept here underpins everything that follows.

2.1 Core Python Concepts

Key Concepts to Learn

- Variables, data types: integers, floats, strings, booleans
- Control flow: if/elif/else statements, for and while loops
- Functions: defining, calling, arguments, return values, scope
- Data structures: lists, tuples, dictionaries, sets
- File I/O: reading and writing CSV and text files
- Error handling: try/except blocks, common exceptions
- List comprehensions and basic functional programming (map, filter)
- Modules and packages: import system, os, sys, math, random
- Object-Oriented Programming basics: classes, objects, methods, inheritance

2.2 First Python Library: NumPy

NumPy is the numerical computing backbone of the entire data science ecosystem. Every other library builds on it.

- Creating and manipulating arrays (1D, 2D, nD)
- Array indexing, slicing, and boolean masking
- Broadcasting rules and vectorized operations
- Statistical functions: mean, median, std, percentile
- Linear algebra basics: dot products, matrix operations
- Random number generation with `numpy.random`

2.3 Data Manipulation: Introduction to Pandas

- Series and DataFrame: creation, indexing, slicing
- Reading data: CSV, Excel, JSON with `pd.read_csv`, `pd.read_excel`
- Basic data exploration: `.head()`, `.info()`, `.describe()`, `.shape`
- Filtering rows and selecting columns
- Handling missing values: `.isna()`, `.dropna()`, `.fillna()`
- Sorting and basic aggregation: `.groupby()`, `.value_counts()`

2.4 Phase 1 Projects

Project	Skills Practiced	Estimated Time
Personal Budget Tracker	Python basics, CSV I/O, lists, dictionaries	1 week
Student Grade Analyzer	NumPy arrays, statistics, functions	1 week
COVID-19 Data Exploration	Pandas basics, filtering, groupby	1–2 weeks
Supermarket Sales Report	Pandas, data cleaning, aggregation	1–2 weeks

2.5 Practical Applications in Phase 1

- Automate repetitive data entry and file management tasks
- Calculate and summarize business metrics from raw CSV exports
- Clean and organize messy datasets for further analysis
- Build simple command-line tools for personal productivity

Milestone check: By the end of Phase 1, you should be comfortable writing Python scripts from scratch, manipulating data in Pandas DataFrames, and completing basic analysis on real datasets.

PHASE 2 — INTERMEDIATE

Phase 2 transforms you from a Python user into a data analyst. You will learn to visualize data persuasively, perform statistical analysis, and build your first machine learning models. This is where data science starts to feel exciting and tangible.

3.1 Advanced Pandas & Data Wrangling

Key Concepts to Learn

- Multi-index DataFrames and hierarchical indexing
- Merging, joining, and concatenating datasets (`.merge()`, `.join()`, `.concat()`)
- Pivot tables and cross-tabulations
- String operations with `.str` accessor and regular expressions
- DateTime handling: parsing, resampling, time-series basics
- Apply, map, and applymap for custom transformations
- Efficient data processing with vectorized operations

3.2 Data Visualization

Matplotlib (Foundation)

- Figure and Axes architecture, subplots
- Line, bar, scatter, histogram, pie charts
- Customization: titles, labels, colors, legends, annotations
- Saving figures as PNG, SVG, PDF

Seaborn (Statistical Visualization)

- Distribution plots: `histplot`, `kdeplot`, `violinplot`, `boxplot`
- Categorical plots: `barplot`, `countplot`, `stripplot`
- Relationship plots: `scatterplot`, `lineplot`, `heatmap`
- `FacetGrid` for multi-panel visualizations
- Statistical annotations and regression lines

Plotly (Interactive Visualization)

- Interactive charts with `plotly.express`
- Dash basics for simple data dashboards

3.3 Statistics for Data Science

- Descriptive statistics: central tendency, spread, skewness, kurtosis
- Probability fundamentals: distributions, the normal curve, CLT

- Hypothesis testing: t-tests, chi-square, ANOVA
- Correlation analysis: Pearson, Spearman
- p-values, confidence intervals, and statistical significance
- A/B testing concepts and experimental design

3.4 Introduction to Machine Learning with Scikit-learn

- The ML pipeline: data split, training, evaluation, prediction
- Supervised learning: Linear Regression, Logistic Regression
- Decision Trees and Random Forests
- K-Nearest Neighbors and Naive Bayes
- Model evaluation: accuracy, precision, recall, F1-score, ROC-AUC
- Cross-validation and overfitting concepts
- Feature engineering: encoding, scaling, creating new features

3.5 Phase 2 Projects

Project	Skills Practiced	Estimated Time
Exploratory Data Analysis on Airbnb Data	Advanced Pandas, Seaborn, Matplotlib	2 weeks
Customer Churn Prediction	Scikit-learn, classification, evaluation metrics	2–3 weeks
House Price Prediction (Regression)	Linear regression, feature engineering, cross-validation	2–3 weeks
Netflix Content Analysis Dashboard	Pandas, Plotly, Seaborn, storytelling with data	1–2 weeks
A/B Test Analysis for E-Commerce	Statistics, hypothesis testing, Pandas	1–2 weeks

3.6 Practical Applications in Phase 2

- Build interactive dashboards for business stakeholders
- Perform customer segmentation and behavioral analysis
- Develop predictive models for business KPIs
- Automate weekly/monthly reporting with Python scripts
- Present data-driven stories using professional visualizations

Milestone check: By the end of Phase 2, you should be able to take a raw dataset, clean and explore it, visualize key insights, and build and evaluate a basic ML model. You are now employable as a Junior Data Analyst.

PHASE 3 — ADVANCED

Phase 3 elevates your skills to a professional data scientist level. You will master advanced machine learning, deep learning foundations, SQL integration, and MLOps basics. Projects in this phase are portfolio-defining.

4.1 Advanced Machine Learning

Ensemble Methods & Boosting

- Gradient Boosting with XGBoost, LightGBM, and CatBoost
- Stacking and blending ensemble techniques
- Hyperparameter tuning: GridSearchCV, RandomizedSearchCV, Optuna

Unsupervised Learning

- Clustering: K-Means, DBSCAN, Hierarchical Clustering
- Dimensionality Reduction: PCA, t-SNE, UMAP
- Anomaly detection techniques

Advanced Feature Engineering

- Feature selection: filter, wrapper, and embedded methods
- Handling imbalanced datasets: SMOTE, class weighting
- Pipelines and ColumnTransformer in Scikit-learn
- Target encoding and advanced encoding strategies

4.2 Deep Learning with TensorFlow & PyTorch

- Neural network fundamentals: neurons, layers, activation functions
- Backpropagation and gradient descent optimizers (Adam, RMSProp)
- Building and training models with Keras (TensorFlow backend)
- Regularization: Dropout, Batch Normalization, L1/L2
- Convolutional Neural Networks (CNNs) for image classification
- Recurrent Neural Networks (RNNs) and LSTMs for sequences
- Transfer learning with pre-trained models (ResNet, EfficientNet)
- Introduction to PyTorch: tensors, autograd, custom datasets

4.3 Natural Language Processing (NLP)

- Text preprocessing: tokenization, stemming, lemmatization, stop words
- Bag of Words, TF-IDF vectorization
- Sentiment analysis with traditional ML and deep learning
- Word embeddings: Word2Vec, GloVe, FastText

- Transformers and BERT with Hugging Face library
- Text classification, named entity recognition (NER)

4.4 SQL & Databases for Data Science

- SQL fundamentals: SELECT, WHERE, GROUP BY, ORDER BY, JOINS
- Advanced SQL: window functions, CTEs, subqueries
- Connecting Python to databases with SQLAlchemy and psycopg2
- Working with PostgreSQL and SQLite
- Introduction to cloud databases: BigQuery, Snowflake basics

4.5 MLOps Fundamentals

- Model versioning and experiment tracking with MLflow
- Model deployment: Flask/FastAPI REST APIs for ML models
- Docker basics for containerizing ML applications
- CI/CD pipelines for ML with GitHub Actions
- Introduction to cloud ML platforms: AWS SageMaker, Google Vertex AI

4.6 Phase 3 Projects

Project	Skills Practiced	Estimated Time
Image Classifier (CNN)	PyTorch/TensorFlow, CNNs, transfer learning	3 weeks
Fraud Detection System	XGBoost, imbalanced data, MLflow tracking	2–3 weeks
Movie Review Sentiment Analyzer (BERT)	Hugging Face, transformers, NLP	3 weeks
End-to-End ML Pipeline with REST API	Scikit-learn, FastAPI, Docker, deployment	3–4 weeks
Time-Series Sales Forecasting	LSTMs, Prophet, XGBoost, cross-validation	2–3 weeks

Milestone check: By the end of Phase 3, you should have 4–5 strong portfolio projects, be comfortable with deep learning, NLP, and model deployment. You are now employable as a Data Scientist or ML Engineer.

PHASE 4 — EXPERT

Phase 4 is about specialization, system design, and leadership in data science. You will master large-scale data engineering, advanced AI techniques, and learn to architect complete data science systems from the ground up.

5.1 Big Data & Cloud Engineering

- Apache Spark with PySpark for distributed data processing
- Data pipelines and ETL with Apache Airflow
- Cloud platforms: AWS (S3, EC2, SageMaker), GCP (BigQuery, Vertex AI)
- Delta Lake and modern data lakehouse architecture
- dbt (data build tool) for SQL-based transformations
- Real-time data streaming with Apache Kafka basics

5.2 Advanced Deep Learning & LLMs

- Generative AI: GANs, VAEs, Diffusion Models
- Large Language Models: fine-tuning with LoRA/QLoRA
- LangChain for building LLM-powered applications
- Retrieval-Augmented Generation (RAG) systems
- Reinforcement Learning: Q-learning, policy gradients, PPO
- Model optimization: quantization, pruning, distillation

5.3 Advanced MLOps & Platform Engineering

- Kubernetes for ML workload orchestration
- Kubeflow and MLflow for full ML lifecycle management
- Feature stores: Feast, Tecton
- Model monitoring: data drift, concept drift, model decay detection
- A/B testing frameworks for ML model updates
- Cost optimization for cloud ML workloads

5.4 Research & Innovation

- Reading and implementing ML papers from arXiv
- Contributing to open-source projects (Scikit-learn, Hugging Face)
- Running rigorous experiments and ablation studies
- Bayesian optimization and AutoML techniques
- Federated learning and privacy-preserving ML

5.5 Phase 4 Capstone Projects

Project	Skills Practiced	Estimated Time
Custom LLM Chatbot with RAG	LangChain, vector DBs, LLM fine-tuning, APIs	4–6 weeks
Real-Time Recommendation Engine	Spark, Kafka, ML serving, cloud deployment	4–6 weeks
End-to-End MLOps Platform	Kubeflow, MLflow, Docker, K8s, monitoring	6–8 weeks
Kaggle Competition (Top 10%)	All skills combined, competitive ML	Ongoing
Open Source Contribution	Collaboration, code quality, documentation	Ongoing

7. Best Resources (Free & Paid)

7.1 Free Resources

Resource	Type	Best For
Python.org Official Tutorial	Documentation	Core Python fundamentals
Kaggle Learn	Interactive Courses	Pandas, ML, Deep Learning, NLP — all free
fast.ai (Practical Deep Learning)	Video Course	Top-down deep learning and PyTorch
StatQuest with Josh Starmer (YouTube)	Videos	Statistics and ML concepts explained visually
Towards Data Science (Medium)	Blog	Applied tutorials and industry insights
Scikit-learn Documentation	Documentation	ML algorithms with excellent examples
Hugging Face Course	Interactive Course	NLP and transformers
Google's ML Crash Course	Course	ML fundamentals with TensorFlow
CS229 Stanford (YouTube)	Lecture Series	In-depth ML theory

7.2 Paid Resources (High ROI)

Resource	Platform	Price Range	Best For
Python for Data Science Bootcamp	Udemy (Jose Portilla)	\$15–20	Comprehensive beginner foundation
Machine Learning Specialization	Coursera (Andrew Ng)	\$49/mo	Gold-standard ML fundamentals
Deep Learning Specialization	Coursera (DeepLearning.AI)	\$49/mo	Neural networks to advanced architectures
Hands-On Machine Learning (Book)	O'Reilly (Geron)	\$50–60	Best ML book with Scikit-learn & TF
Python Data Science Handbook (Book)	O'Reilly (VanderPlas)	\$50–60	NumPy, Pandas, Matplotlib, Scikit-learn
DataCamp Career Tracks	DataCamp	\$25/mo	Structured, skill-specific tracks
Zero to Mastery — Data Science	ZTM Academy	\$23/mo	Project-based learning

7.3 Practice & Competition Platforms

- Kaggle — datasets, competitions, and free GPU notebooks
- LeetCode — Python coding challenges for interviews
- StrataScratch — SQL and Python interview questions from real companies
- DrivenData — mission-driven data science competitions
- Alcrowd — AI research competitions and challenges

8. Common Mistakes to Avoid

8.1 Learning Mistakes

- **Tutorial hell:** Watching course after course without building anything. Fix: for every 2 hours of learning, spend 2 hours coding.
- **Skipping fundamentals:** Jumping to deep learning before mastering Pandas and statistics. Fix: follow this roadmap's phases in order.
- **Memorizing instead of understanding:** Copying code without knowing why it works. Fix: always type code manually and experiment with changes.
- **Ignoring mathematics:** Treating ML as a black box. Fix: study the intuition behind algorithms — you don't need a PhD, but you need the basics.
- **Learning multiple tools simultaneously:** Trying TensorFlow, PyTorch, and Keras at once. Fix: master one tool deeply before branching out.

8.2 Project & Portfolio Mistakes

- **Only using Titanic and Iris datasets:** These are overused and do not impress employers. Fix: find real-world datasets from Kaggle, government open data, or APIs.
- **No documentation:** Submitting notebooks with no explanations. Fix: write markdown cells explaining your thought process and findings.
- **Skipping model evaluation:** Reporting only accuracy. Fix: always use multiple metrics — confusion matrix, F1, ROC-AUC, and explain why each matters.
- **Not cleaning data properly:** Training models on raw uncleaned data. Fix: invest 60–70% of project time in EDA and data preparation.

8.3 Career Mistakes

- **Waiting until you are 'ready':** There is no perfect moment. Apply for jobs when you have 3–4 strong projects completed.
- **Ignoring networking:** 90% of jobs are filled through connections. Fix: engage actively on LinkedIn, attend meetups, join data science communities.
- **Not practicing communication:** Technical skill alone is not enough. Fix: practice explaining your projects to non-technical friends and family.

9. Building a Strong Data Science Portfolio

Your portfolio is your most powerful job application tool. A well-structured GitHub profile with 5–6 strong projects will outperform a resume alone in almost every hiring scenario.

9.1 Portfolio Structure

Project Type	Quantity	Purpose
EDA + Storytelling Project	1–2	Show data intuition and communication skills
Machine Learning Project (Regression or Classification)	1–2	Core data science competency
Deep Learning or NLP Project	1	Signal advanced technical capability
End-to-End Deployed Application	1	Shows production-readiness (FastAPI + Docker + cloud)
Domain-Specific Project (Finance, Health, etc.)	1	Demonstrates industry knowledge and passion

9.2 Making Each Project Stand Out

- Use real-world datasets — not only classic Kaggle beginner sets
- Write a clear README with problem statement, approach, results, and how to run the project
- Include a short video walkthrough (Loom) for your best projects
- Add business interpretation: 'This model would save \$X or improve Y by Z%'
- Show iterative improvement — document what you tried and what worked
- Deploy at least one project live using Streamlit, Hugging Face Spaces, or Railway

9.3 Other Portfolio Channels

- LinkedIn: Share project posts with visuals and brief insights — consistency matters more than perfection
- Medium / Substack: Write about what you learned — teaching solidifies understanding
- Kaggle Profile: A public Kaggle Expert or Master rank is a strong credibility signal
- GitHub: Keep repositories clean, well-documented, and regularly updated

10. Career Transition Guide

10.1 Data Analyst

Timeline from beginner: 4–6 months | Salary range (India): ₹4–12 LPA | Salary range (Global): \$55,000–\$90,000

Core Requirements

- Strong Python and Pandas skills
- SQL proficiency (critical — practice on StrataScratch)
- Excellent data visualization skills (Tableau or Power BI is a bonus)
- Clear communication and business acumen
- 2–3 EDA portfolio projects on business datasets

Recommended Learning Path

- Complete Phases 1 and 2 of this roadmap
- Learn SQL via Mode Analytics or StrataScratch
- Build a Tableau or Power BI dashboard on a business dataset
- Practice translating analysis into business recommendations

10.2 Data Scientist

Timeline from beginner: 9–14 months | Salary range (India): ₹8–25 LPA | Salary range (Global): \$85,000–\$140,000

Core Requirements

- All Data Analyst skills plus strong ML foundations
- Experience with the complete model lifecycle: EDA, feature engineering, training, evaluation, deployment
- Statistics knowledge: hypothesis testing, distributions, A/B testing
- Communication of model results to non-technical stakeholders
- 4–5 diverse portfolio projects including at least one deployed model

Recommended Learning Path

- Complete Phases 1 through 3 of this roadmap
- Participate in at least 2 Kaggle competitions
- Deploy one project as a REST API (FastAPI + Docker)
- Practice DS interviews on LeetCode (medium Python) and case-study platforms

10.3 Machine Learning Engineer

Timeline from beginner: 14–20 months | Salary range (India): ₹12–40 LPA | Salary range (Global): \$110,000–\$200,000

Core Requirements

- Strong software engineering skills: clean code, testing, design patterns
- Deep ML knowledge plus MLOps (MLflow, Docker, Kubernetes, CI/CD)
- Cloud platform experience (AWS, GCP, or Azure)
- System design for ML: feature stores, model serving, monitoring
- Experience deploying models that serve millions of predictions in production

Recommended Learning Path

- Complete all 4 phases of this roadmap
- Build an end-to-end MLOps pipeline as a capstone project
- Study system design for ML interviews (Chip Huyen's 'Designing Machine Learning Systems')
- Obtain at least one cloud certification (AWS ML Specialty or GCP Professional ML Engineer)

10.4 Interview Preparation Timeline

Stage	Focus Areas	Resources
3 months before applying	Identify target roles and companies; assess skills gaps	LinkedIn Jobs, Glassdoor, Levels.fyi
2 months before applying	LeetCode Python (Easy–Medium); SQL practice; ML theory review	LeetCode, StrataScratch, StatQuest
1 month before applying	Mock interviews; refine portfolio; tailor resume per role	Pramp, Interviewing.io, resumeworded.com
Applying	Send 10–15 targeted applications per week; prioritize roles matching 70–80% of requirements	LinkedIn, Naukri, AngelList
Interviewing	System design (MLE), case studies (DS/DA), behavioral stories (all roles)	Chip Huyen ML Systems, own project stories

Final Words of Encouragement

The path from beginner to expert in data science is a marathon, not a sprint. Every professional working in the field today started exactly where you are right now. The difference between those who make it and those who give up is not talent — it is consistency, curiosity, and resilience.

Build projects you are genuinely curious about. Write about what you learn. Connect with the community. Apply before you feel ready. The data science field is vast, rewarding, and growing — and there is a place in it for you.

Good luck. Now go write some code.
